

KELVIN WARUI KIONDO

Mechatronics Engineer · Industrial Automation, Instrumentation & Energy Systems Monitoring

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PROFESSIONAL PROFILE

Mechatronics Engineer (B.Sc., Dedan Kimathi University of Technology) working at the intersection of industrial plant, instrumentation and software. My work has covered three areas that rarely sit in one CV: factory-floor automation and electrical systems (PLC programming, ladder logic, packaging-line automation, industrial rewiring, refrigeration and ventilation plant); measurement and control engineering (sensor fusion, state estimation, LQR controller design, custom telemetry and data-logging software); and full-stack systems development (real-time monitoring dashboards, time-series databases, automated reporting engines, secure client portals).

That combination means I can do more than read a plant — I can instrument it, log it, and build the reporting layer that turns raw meter and sensor output into a live, auditable dataset. My "SNC Horizon" project is a working example: a SCADA-style digital twin that ingests continuous telemetry and renders it as an interactive real-time dashboard. I am currently a part-time Technical Trainer at Kiambu Institute of Science and Technology, where I teach PLCs, ladder logic, hydraulics and pneumatics, and run the Mechatronics laboratory.

I am looking to apply this to energy management, measurement and verification, and industrial efficiency work — where the bottleneck is very often not the engineering judgement but the metering, data capture and reporting infrastructure needed to prove savings.

TECHNICAL COMPETENCIES

- **Industrial Automation & Control:** PLC programming and ladder logic (Siemens); SCADA concepts and interface design; ISA S88 / S95 batch and enterprise-control standards; packaging-line automation; hydraulics and pneumatics; process descriptions and P&ID documentation.
- **Electrical Systems:** Building and industrial wiring (first-fix and second-fix); conduit, cabling and distribution board installation; electrical load calculation and demand assessment; switchgear selection; industrial rewiring and overhaul; testing and safety verification using multimeters and insulation testers.
- **Instrumentation & Measurement:** Sensor selection and integration (IMU, GPS, thermal, optical); Madgwick sensor-fusion filter implementation in C; multi-channel data logging; custom telemetry link design; calibration and signal integrity.
- **HVAC & Thermal Plant:** Ventilation ducting layout and installation; cooling-unit installation and electrical control integration; maintenance and optimisation of large-scale industrial cooling and ventilation systems for temperature-sensitive production.
- **Modelling & Simulation:** MATLAB / Simulink (6-DOF dynamic modelling, state-space analysis); Linear Quadratic Regulator (LQR) and PID controller design; ANSYS finite-element analysis; Altair FluxMotor for electrical machine design and performance characterisation.
- **Design & Fabrication:** SolidWorks, Autodesk Inventor, AutoCAD, Onshape, OpenSCAD; KiCad PCB and schematic design; CNC machining and lathe work; mechanical fabrication and enclosure design.
- **Data Systems & Software:** PostgreSQL / Supabase, Google Firebase; Next.js 14 (React), Flutter / FlutterFlow, Vercel; C, C++, C#, Python; REST API design and integration; server-side cron scheduling for automated reporting; row-level security, AES-256 at rest and TLS 1.3 in transit; Linux system administration.
- **Materials & Documentation:** Bill of Materials preparation for electrical and automation packages; SAP materials coordination; quantity take-off and cost estimation from architectural and engineering drawings; as-built documentation; software design documents and end-user test protocols.
- **Professional:** Technical training and curriculum delivery; laboratory management; team supervision; client-facing consulting; technical report writing; English and Kiswahili.

ENGINEERING & TECHNICAL EXPERIENCE

Technical Trainer & Laboratory Systems Administrator (Part-time) | Kiambu Institute of Science and Technology (KIST)

Kiambu, Kenya · February 2026 – Present

- **Technical instruction:** Deliver and assess practical training modules in PLC programming, ladder diagram logic, hydraulics and pneumatics to TVET-level students, translating industrial control theory into hands-on workshop competence.
- **Laboratory management:** Maintain and optimise the physical and digital infrastructure of the Mechatronics laboratory, including test benches, trainer rigs and instrumentation, ensuring operational readiness for continuous student and project use.
- **KINAP Lab inventory system:** Architected and deployed a hybrid digital inventory and point-of-sale system for the Mechatronics laboratory, replacing manual paper tracking of components and consumables with an automated, queryable database — giving the department accurate, real-time visibility of stock and usage for the first time.
- **Assessment design:** Developed evaluation rubrics and structured practical assessments for automation and control modules.

Independent Systems Engineer & Technical Consultant | Self-employed / Consulting

Kenya and remote clients (USA) · July 2023 – Present

- **Monitoring & telemetry systems:** Design and build real-time monitoring platforms that ingest continuous machine and infrastructure telemetry and present it through interactive dashboards — including topology visualisation, live load and latency tracking, and threshold alerting.
- **Database architecture:** Design and administer PostgreSQL (Supabase) and Firebase backends for operational data, including time-stamped event logging, role-based access control and row-level security policies.
- **Automated reporting engines:** Implement server-side scheduled jobs that compile live database state into formatted, printable reports and multi-stage advance notifications — removing manual report assembly from the client's workflow entirely.
- **Requirements & documentation:** Produce software design documents, system architecture specifications and end-user acceptance test protocols; run client onboarding and technical training on delivered systems.
- **Web infrastructure:** Rebuild and maintain client web platforms, including a full front-end rebuild and third-party data integration (Georgia Multiple Listing Service) for a US real-estate client.

Lead Electrical & Building Systems Contractor | Independent Projects

Ruiru and Nairobi, Kenya · May 2024 – Present

- **Building electrical installation:** Planned and executed complete electrical installations for residential and commercial projects — full circuit layout from main intake through distribution boards to final socket, switch and luminaire fitting, across both first-fix and second-fix stages.
- **Load calculation & sizing:** Calculated building electrical demand to size conductors, protective devices and distribution equipment correctly, ensuring installations remain within safe thermal and capacity limits under sustained load.
- **HVAC installation & integration:** Installed ventilation ducting and cooling plant for commercial office space, and wired and commissioned the electrical control systems driving them — ensuring mechanical plant (compressors, pumps, fans) was correctly integrated with building power and control.
- **Quantity take-off & estimation:** Produced detailed material schedules (cable lengths, conduit, switchgear, HVAC components) from architectural drawings for construction managers, enabling accurate supplier quotation and minimising material waste.
- **Site supervision:** Supervised technician teams on site, verifying compliance with wiring safety codes and conformity of mechanical installations to structural drawings.

Industrial Attaché — Embedded Systems & Instrumentation | Atego School (Beyond Grades)

Kenya · February 2022 – April 2022

- **PCB design:** Designed and fabricated custom printed circuit boards in KiCad, from schematic capture through layout to board bring-up and testing.
- **Embedded development:** Programmed microcontroller firmware for several working sensor-actuator systems, including an automated tap (proximity-sensing water control), a fire detection and alarm system, and an automated page-turning mechanism.
- **Safety systems installation:** Planned and installed building-wide fire detection wiring and sensor placement — low-voltage system design, cable routing and device siting.
- **Documentation & procurement:** Produced as-built drawings recording exact wiring and component locations for future maintenance; developed IT hardware standards; prepared Bills of Materials for electrical and automation solutions; acted as technical lead on component selection and supplier evaluation.
- **Verification:** Built virtual test environments and wrote end-user test protocols for delivered hardware.

Industrial Attaché — Factory Automation & Electrical | New Kenya Cooperative Creameries (NKCC)

- **PLC programming:** Programmed and modified factory PLCs controlling production plant, working to ISA S88 (batch control) and S95 (enterprise-control integration) standards.
- **Automation installation:** Installed automation systems — both hardware and control software — for milk packaging machinery, including sensor and actuator commissioning on the line.
- **Electrical overhaul:** Participated in a full overhaul of the plant's electrical wiring, replacing ageing distribution infrastructure with modern, higher-capacity systems and testing all new connections against national safety standards.
- **Refrigeration & ventilation plant:** Worked on the plant's large-scale industrial cooling and ventilation systems — the dominant energy load in a dairy facility — with responsibility for keeping mechanical plant running efficiently to protect temperature-sensitive product.
- **BOM & SAP materials control:** Prepared Bills of Materials for electrical and automation packages and coordinated electrical and automation items through SAP, maintaining full accountability for every component consumed during the overhaul.

SELECTED ENGINEERING PROJECTS

Self-Piloting Octocopter UAV — *Final year engineering project, DeKUT · May 2021 – January 2022*

Design, fabrication and flight testing of an eight-rotor autonomous aerial vehicle capable of operating under radio control or fully autonomously via an onboard GPRS-linked navigation system. The airframe achieved three qualified capabilities: sustained position hold at a commanded altitude, latitude and longitude under wind disturbance; autonomous return-to-home on task completion or fault; and autonomous waypoint tracking along a pre-programmed flight plan without pilot input.

- **Dynamic modelling:** Derived and simulated the full 6-degree-of-freedom equations of motion in MATLAB and Simulink to characterise the airframe's open-loop behaviour before hardware commitment.
- **Optimal control design:** Designed a Linear Quadratic Regulator to stabilise and command the vehicle's twelve-state model, tuning the cost weightings against simulated disturbance response.
- **Sensor fusion:** Implemented the Madgwick attitude-estimation filter in C, fusing gyroscope, accelerometer and magnetometer data into a stable, drift-corrected orientation estimate on resource-constrained embedded hardware.
- **Telemetry & data logging:** Wrote custom ground-station software in C# for live remote piloting, continuous telemetry capture and post-flight data analysis.
- **Fabrication:** Machined and assembled the airframe and integrated the full power, propulsion, sensing and communications stack.

"SNC Horizon" — **Infrastructure Digital Twin & Telemetry Monitor** — *Independent consulting project*

A real-time network topology visualiser and infrastructure monitoring dashboard. Raw continuous telemetry — latency, CPU load, security handshake events — is ingested and rendered through custom mathematical drawing routines (Flutter CanvasKit) into a live, interactive SCADA-style interface, giving operators an at-a-glance view of system state and the ability to drill into any node. The same architecture — continuous data acquisition, time-series storage, live graphical presentation and thresholded alerting — is directly transferable to metered energy and utility monitoring.

Muchemi Law Practice Management Portal — *Independent consulting project*

A secure, full-stack legal practice management system built on Next.js 14 with a Supabase (PostgreSQL) backend, deployed with paired web and Android clients.

- **Automated notification engine:** Server-side cron jobs generate a staged alert cascade — three days, one day and one hour ahead of every scheduled event — eliminating manual diary chasing.
- **Zero-trust security model:** AES-256 encryption at rest, TLS 1.3 in transit and database-level Row Level Security enforcing per-user data isolation.
- **One-click document generation:** A dynamic docket generator that compiles live database state into formatted, print-ready PDFs on demand.

iLand Loyalty — **Digital Rewards & Merchant Platform** — *Independent consulting project*

A cloud-based customer loyalty and retention platform pairing a customer-facing digital wallet with a staff scanning terminal, synchronised in real time through a secure Firebase backend and supported by a merchant analytics dashboard. Involved complex client-side state management (Riverpod), Stripe payment integration, and end-to-end management of the merchant onboarding and data-entry workflow.

Chartzy — **Clinical Scheduling & Records Application** — *Waki Software Solutions*

A management application tracking medication schedules, appointment calendars and patient history, built on a real-time database backend with automated reminder alerts. The underlying scheduling-and-history data model is the same pattern used for asset maintenance scheduling and service-interval tracking.

KINAP Laboratory Inventory & POS System — *Kiambu Institute of Science and Technology*

A hybrid online/offline inventory management and point-of-sale system for the KIST Mechatronics laboratory, replacing manual component tracking with an automated digital solution that survives intermittent connectivity and reconciles on reconnection.

EDUCATION

B.Sc. Mechatronics Engineering | Dedan Kimathi University of Technology (DeKUT), Nyeri

August 2016 – May 2022

- **Specialisation:** Automotive Engineering (final year elective stream) — internal combustion engines, vehicle dynamics, powertrain and transmission logic.
- **Core coursework:** Modern control systems, electrical machines and drives, thermodynamics and fluid mechanics, industrial automation and instrumentation, engineering materials, mechanical design, embedded systems, engineering simulation and analysis.
- **Final year project:** Design, fabrication and testing of a self-piloting octocopter UAV (detailed above).

Kenya Certificate of Secondary Education — Grade A- | St. Mary's Boys Academy, Nyeri

2012 – 2015

Kenya Certificate of Primary Education — Grade A | St. Lwanga Primary School, Naro Moru

2004 – 2012

CERTIFICATIONS & PROFESSIONAL DEVELOPMENT

- No-Code Operator Course — rapid deployment and management of scalable digital tools and automated business workflows (2024).
- Fundamentals of Digital Marketing — Google Digital Skills for Africa.
- Communications Foundations — LinkedIn Learning.
- Time Management Foundations — LinkedIn Learning.
- Essentials of Team Collaboration — LinkedIn Learning.
- Mastering In-Demand Soft Skills — LinkedIn Learning.
- Virtual Assistant & Remote Operations Training — Runnovate (2021).

AWARDS & RECOGNITION

- **The Rockefeller Foundation – Acumen Student Social Innovation Challenge, Global Winner (2020).** Won as part of the Runnovate team, competing against student-run social enterprises worldwide.
- **Mid-Atlantic #1 ranked positive-content streaming network (2022).** Contributed as digital lead to Mogul TV Global Network's expansion to coverage in over 30 countries and its top ranking in category.

ADDITIONAL PROFESSIONAL EXPERIENCE — OPERATIONS, TRAINING & DIGITAL

Included for completeness. This period ran alongside and between my engineering roles, and is where I built the client-facing, training and team-leadership side of my work.

Digital Specialist | Louis E. Page Inc., Massachusetts, USA (remote)

August 2022 – April 2023

- Maintained and developed the company website, resolving faults and publishing new content.
- Designed and executed a recurring email marketing programme to a targeted customer list; company sales doubled over the engagement.
- Produced and scheduled marketing content across Facebook, Instagram and LinkedIn.

Digital Marketing Specialist | Mogul TV Global Network, Virginia, USA (remote)

May 2022 – November 2022

- Managed social media strategy, SEO, lead generation and project coordination for an international streaming network.

- Grew platform following by 10% with a corresponding rise in visibility; researched, drafted, edited and distributed blog content supporting the network's expansion to over 30 countries.

Digital Marketing Skills Trainer | Runnovate

March 2021 – August 2021

- Designed and delivered digital marketing training to cohorts of learners — my first formal instructional role, and the foundation for the technical training work I now do at KIST.
- Researched, drafted, edited and distributed long-form content to target audiences.

Director of Operations | Runnovate

March 2020 – October 2020

- Oversaw all active company projects and training programmes, and set growth strategy including accelerator programme applications and participation.
- Led the team to victory in the 2020 Rockefeller Foundation – Acumen Student Social Innovation Challenge.

Web Developer & Social Media Manager | Brachel Jones Realty, Georgia, USA (remote)

July 2023 – Present

- Rebuilt the company website — new user interface, improved navigation and load performance — and integrated it with the Georgia Multiple Listing Service (MLS) feed.
- Retained on ongoing maintenance; manage social channels and build targeted listing advertisement campaigns.

REFEREES

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